

10 Moment hybnosti

$$\vec{L} = \vec{r} \times \vec{p}$$

$$[L_i, L_j] = i\hbar \sum_k \epsilon_{ijk} L_k$$

$$L^2 = L_x^2 + L_y^2 + L_z^2, \quad [L^2, L_i] = 0$$

- předpokládáme, že máme řešení L^2 a L_z :

3.1 Kvantování momentu hybnosti

→ důležitý problém => máme 2 kvantová čísla λ, m

$$L^2 |\lambda, m\rangle = \lambda \hbar^2 |\lambda, m\rangle \quad L_z |\lambda, m\rangle = m \hbar |\lambda, m\rangle$$

1. přípravný bod

=> analogicky anihilující a kvantující operátory:

$$L_+ := L_x + iL_y \quad L_- := L_x - iL_y$$

$$L_+^* = L_- \quad L_-^* = L_+$$

$$[L_z, L_-] = [L_z, L_x] - i[L_z, L_y] =$$

$$= L_y i\hbar - i(-i\hbar L_x) = -\hbar L_-$$

$$[L_z, L_+] = \hbar L_+$$

$$[L^2, L_{\pm}] = 0$$

$$L_- L_+ = (L_x - iL_y)(L_x + iL_y) = L_x^2 + L_y^2 + i(L_x L_y - L_y L_x)$$

$$= L_x^2 + L_y^2 + i(\hbar)L_z = L^2 - L_z^2 - \hbar L_z$$

$$L_+ L_- = L^2 - L_z^2 + \hbar L_z$$

2. přípravný bod

$$L^{\pm} (L^{\pm} |\lambda, m\rangle) = L^{\pm} (L^2 |\lambda, m\rangle) = L^{\pm} (\lambda \hbar^2 |\lambda, m\rangle)$$

$$= \lambda \hbar^2 L^{\pm} |\lambda, m\rangle$$

$$L_z (L_+ |\lambda, m\rangle) = (\hbar L_+ + L_+ L_z) |\lambda, m\rangle$$

$$= \hbar(m+1) L_+ |\lambda, m\rangle$$

$$L_z (L_- |\lambda, m\rangle) = \hbar(m-1) L_- |\lambda, m\rangle$$

3. přípravný bod

4 vzhledy:

$$\langle 2 | L_- L_+ | 2 \rangle = \langle L_+ 2 | L_+ 2 \rangle = \|L_+ 2\|^2 \geq 0$$

$$\langle 2 | L_+ L_- | 2 \rangle \geq 0 \quad \text{analogicky}$$

$$\frac{1}{2} (L_+ L_- + L_- L_+) = L^2 - L_z^2$$

$$2 = |\lambda, m\rangle \quad 0 \leq \langle \lambda, m | \frac{1}{2} (L_+ L_- + L_- L_+) | \lambda, m \rangle$$

$$= \langle \lambda, m | L^2 - L_z^2 | \lambda, m \rangle = \lambda \hbar^2 - m^2 \hbar^2$$

$$\Rightarrow \lambda \geq m^2 \quad \wedge \quad \lambda \geq 0$$

$$\sqrt{\lambda} \geq |m|$$

$$\lambda \text{ celé} \Rightarrow \lambda = m^2 \text{ nebo } m^2 + 1 \quad \lambda' = m^2 \text{ nebo } m^2 + 1$$

$$\rightarrow \lambda \geq \lambda' \quad | \lambda, \lambda \rangle \neq 0 \quad \wedge \quad L_+ | \lambda, \lambda \rangle = 0$$

$$| \lambda, \lambda' \rangle \neq 0 \quad \wedge \quad L_- | \lambda, \lambda' \rangle = 0$$

$\sim | \lambda, \lambda+1 \rangle$
2+1 by bylo větší vlastně číslo
š s tím, že
2 je nejmenší

4. přípravný bod

$$0 = L_- L_+ | \lambda, \lambda \rangle = (L^2 - L_z^2 - \hbar L_z) | \lambda, \lambda \rangle$$

$$= (\lambda \hbar^2 - \lambda^2 \hbar^2 - \lambda \hbar^2) | \lambda, \lambda \rangle$$

$$0 = L_+ L_- | \lambda, \lambda' \rangle = (L^2 - L_z^2 + \hbar L_z) | \lambda, \lambda' \rangle$$

$$= (\lambda \hbar^2 - \lambda'^2 \hbar^2 + \lambda' \hbar^2) | \lambda, \lambda' \rangle$$

$$\Rightarrow \lambda - \lambda'^2 - \lambda = 0$$

$$\Rightarrow \lambda' = \lambda + 1 \quad \text{společně definice}$$

$$\lambda - \lambda'^2 + \lambda' = 0$$

$$\lambda' = -\lambda$$

$$\lambda = \lambda(\lambda+1)$$

$$\Rightarrow | \lambda, m \rangle \rightarrow | \lambda, m \rangle$$

5. přípravný bod

$$L_- | \lambda, \lambda \rangle \sim | \lambda, \lambda-1 \rangle$$

$$L_-^s | \lambda, \lambda \rangle \sim | \lambda, \lambda-s \rangle$$

$$\rightarrow \lambda-s = -\lambda \Rightarrow s = 2\lambda \in \mathbb{N}_0$$

$$\Rightarrow \lambda = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$$

$$L^2 \quad \lambda(\lambda+1)\hbar^2 \quad \lambda = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$$

$$L_z \quad 2\lambda+1 \text{ hodnot: } m = \lambda, \lambda-1, \dots, -\lambda+1, -\lambda$$

obecně:

$$L_+ | \lambda, m \rangle = c_{\lambda m}^+ | \lambda, m+1 \rangle$$

$$L_- | \lambda, m \rangle = c_{\lambda m}^- | \lambda, m-1 \rangle$$

$$\langle \lambda, m | L_- L_+ | \lambda, m \rangle = \|L_+ | \lambda, m \rangle\|^2 = |c_{\lambda m}^+|^2$$

$$\langle \lambda, m | L^2 - L_z^2 - \hbar L_z | \lambda, m \rangle = \hbar^2 [\lambda(\lambda+1) - m^2 - m]$$

$$\text{BÚNO ai m fci: } c_{\lambda m}^+ = \hbar \sqrt{\lambda(\lambda+1) - m^2 - m}$$

$$= \hbar \sqrt{(\lambda - m)(\lambda + m + 1)}$$

anal.:

$$c_{\lambda m}^- = \hbar \sqrt{\lambda(\lambda+1) - m^2 + m}$$

$$= \hbar \sqrt{(\lambda + m)(\lambda - m + 1)}$$

3.2 Maticová reprezentace

$$\beta = \{ | \lambda, m \rangle : \lambda = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots ; m = \lambda, \dots, -\lambda \}$$

$$(L^2)_{\beta} = \hbar^2 \begin{pmatrix} 0 & & & 0 \\ \frac{3}{4} & & & \\ & \frac{1}{4} & & \\ 0 & & \ddots & \end{pmatrix}$$

$$(L_z)_{\beta} = \begin{pmatrix} 0 & \frac{1}{2} & & 0 \\ & \frac{1}{2} & & \\ 0 & & \ddots & \\ & & & -\frac{1}{2} \end{pmatrix} \hbar$$

$$\langle \lambda', m' | L_+ | \lambda, m \rangle = \langle \lambda', m' | c_{\lambda m}^+ | \lambda, m+1 \rangle$$

$$= c_{\lambda m}^+ \delta_{\lambda \lambda'} \delta_{(m+1)m'}$$

$$L_x = \frac{1}{2} (L_+ + L_-) \quad L_y = \frac{1}{2i} (L_+ - L_-)$$

$$| \lambda, m \rangle : | \lambda, \lambda \rangle, | \lambda, \lambda-1 \rangle, \dots, | \lambda, -\lambda \rangle$$

$$\lambda=0 \quad \|L_z\| = \|L_+\| = \|L_-\| = \|L_x\| = \|L_y\| = \|L^2\| = 0$$

$$\lambda = \frac{1}{2} \rightarrow | \frac{1}{2}, \frac{1}{2} \rangle, | \frac{1}{2}, -\frac{1}{2} \rangle$$

$$\Rightarrow c_{\frac{1}{2} \frac{1}{2}}^+ = 0 \quad c_{\frac{1}{2} -\frac{1}{2}}^+ = \hbar$$

$$\langle \frac{1}{2}, \frac{1}{2} | L_+ | \frac{1}{2}, \frac{1}{2} \rangle = 0$$

$$\langle \frac{1}{2}, \frac{1}{2} | L_x | \frac{1}{2}, -\frac{1}{2} \rangle = \hbar$$

$$\langle \frac{1}{2}, -\frac{1}{2} | L_+ | \frac{1}{2}, \frac{1}{2} \rangle = 0$$

$$\langle -\frac{1}{2}, -\frac{1}{2} | L_+ | -\frac{1}{2}, -\frac{1}{2} \rangle = 0$$

$$\Rightarrow \|L_-\| \text{ sdružený operátor k } \|L_+\| \Rightarrow \|L_-\| = \hbar \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

$$\Rightarrow \|L_x\| = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \|L_y\| = \frac{\hbar}{2i} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\|L_z\| = \hbar \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

→ Pauliho matice: $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ σ_z je diagonální

$$\Rightarrow \|L_x\| = \frac{\hbar}{2} \sigma_x \quad \|L_y\| = \frac{\hbar}{2} \sigma_y \quad \|L_z\| = \frac{\hbar}{2} \sigma_z$$

10.3 Vlastní funkce operátoru orbitálního L

$$L^2 | \lambda, m \rangle = \lambda(\lambda+1) \hbar^2 | \lambda, m \rangle \quad L_z | \lambda, m \rangle = m \hbar | \lambda, m \rangle$$

$$| \lambda, m \rangle = Y_{\lambda m}(\theta, \varphi) = N_{\lambda m} P_{\lambda}^{(m)}(\cos \theta) e^{im\varphi}$$

$$N_{\lambda m} = \sqrt{\frac{(\lambda - |m|)! (2\lambda + 1)}{(\lambda + |m|)! 4\pi}}$$

$$P_{\lambda}^{(m)} \text{ je prodloužený Legendrovův polynom}$$

$$P_{\lambda}(\xi) = \frac{1}{2^{\lambda} \lambda!} \frac{d^{\lambda}}{d\xi^{\lambda}} (\xi^2 - 1)^{\lambda}$$

$$P_{\lambda}^{(m)}(\xi) = (1 - \xi^2)^{\frac{|m|}{2}} \frac{d^{|m|}}{d\xi^{|m|}} P_{\lambda}(\xi)$$

$$P_{\lambda}^0(\xi) = P_{\lambda}(\xi)$$

$$P_0(\xi) = 1 \quad P_1(\xi) = \xi$$

$$P_1^1(\xi) = \sqrt{1 - \xi^2}$$

$$Y_0^0(\theta, \varphi) = \frac{1}{\sqrt{4\pi}} \quad Y_1^0(\theta, \varphi) = \sqrt{\frac{3}{4\pi}} \cos \theta$$

$$Y_1^1(\theta, \varphi) = \sqrt{\frac{3}{8\pi}} \sin \theta e^{i\varphi} \quad Y_1^{-1}(\theta, \varphi) = \sqrt{\frac{3}{8\pi}} \sin \theta e^{-i\varphi}$$